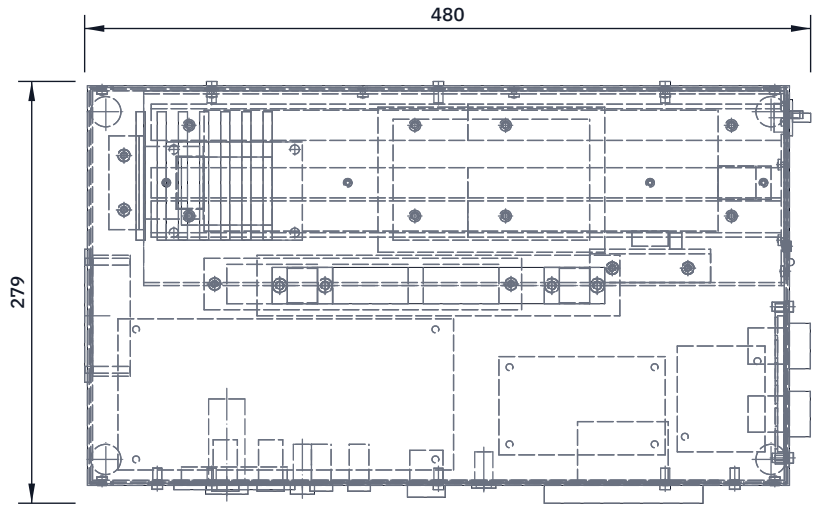


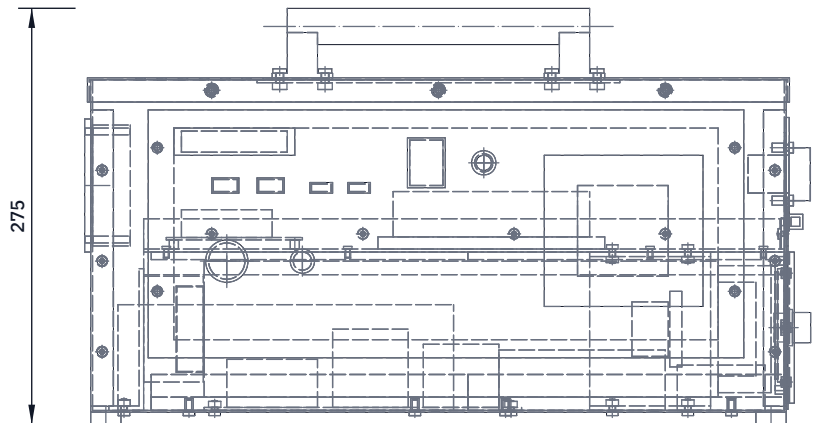
CONCEPT, NOT FOR FABRICATION

PowerBox · github.com/BoujeeEnjinia1701/powerbox



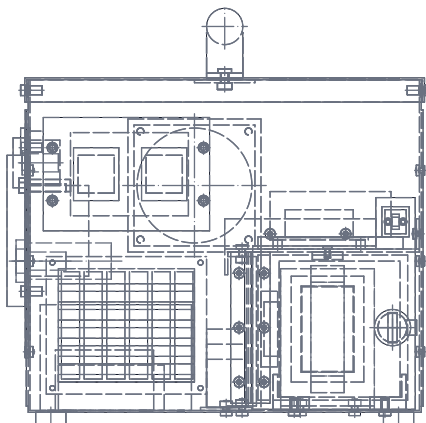
TOP VIEW

Scale 1:5; looking down (along -Z)



FRONT VIEW

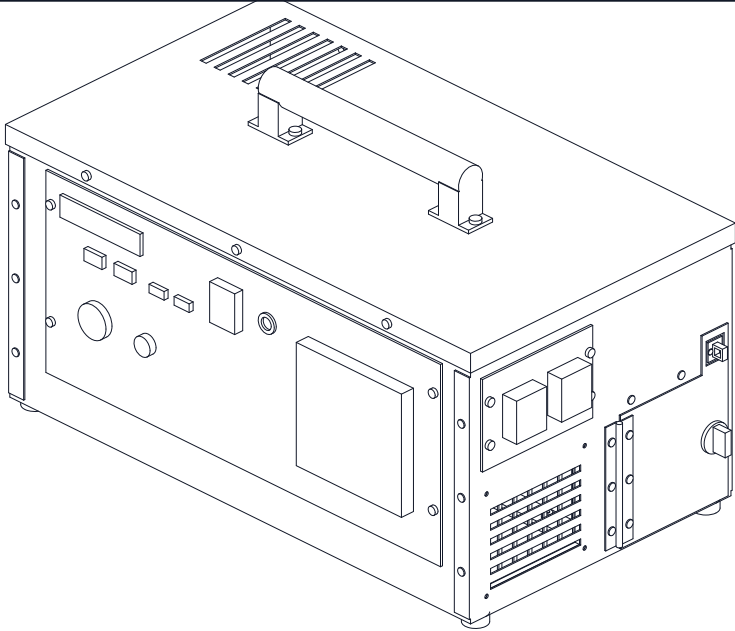
Scale 1:5; looking at the front (along +Y)



RIGHT VIEW

Scale 1:5; looking at the right side (along -X)

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA, SwapCell interface v0.3 (PBX-CAL-001)	2026-09-25	AC
P2	Constructable design, PBX-DDR-003	2026-10-01	AC
P3	Lit rocker switch cut-out 22 x 30; R2 and decisions of 2026-10-02	2026-10-02	AC



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS AND INTERFACES (MM)

Body 460 x 260 x 220; overall 480 x 279 x 275
Sheet 1.2 Al; lid skirt 16 over the walls, 6 M4 screws
Handle span 200 on a 2 mm doubler; 8 mm rubber feet
Pack bay: SwapCell interface v0.3, pack 340 x 90 x 80, 393 overall; inserted from +X end, connector first
Printed runners and top rail, 1 clearance per side
Class D catch on a folded 3 mm bracket; 2 mm folded shelf
Door hinged on its front edge; cam latch; padlock hasp
Receptacle: floating mount, 10 kOhm INTERLOCK coding resistor (v0.3 item W); 120 ohm CAN termination
Station host type 3, mode 4 charge-discharge, 5.0 A max
Front: USB-C 100 + 60 W, 2 x USB-A, 2 x 12 V, lit switch, 230 V RCD
+X end: DC in 12 to 60 V 200 W, charger in 54.6 V 5 A
Main fuse 30 A, inverter breaker 20 A (PBX-CAL-001)
Mass about 9.4 kg with pack (PBX-CAL-001 v0.4)
PRELIMINARY, NOT FOR FABRICATION

POWERBOX

github.com/BoujeeEnjinia1701/powerbox

General arrangement, SwapCell station

DRAWING NO. PBX-DWG-001			REV P3	SHEET 1 of 1	SIZE ANSI B
SCALE 1:5	UNITS mm	PROJECTION 	DATE 2026-10-02		DRAWN Amish Chadha
MATERIAL / NOTES Case, lid, door: 1.2 mm 5052 Al; shelf 2 mm, brackets 3 mm Al; runners and rail printed PETG. See bom/bom.csv					
Licensed CERN-OHL-S-2.0 · Design Molecule Lab · designmolecule.com				Generated from cad/src; do not edit by hand	